

TO DANG MINH TUAN

Software Engineer · Java · Spring Boot · AWS · 3rd-Year @ UIT

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PROFESSIONAL SUMMARY

3rd-year Information Systems student (UIT) with 2+ production-grade projects in Java/Spring Boot backend development, RESTful API design, and cloud deployment (AWS). Engineered concurrency-safe systems using Redis + JPA optimistic locking, automated end-to-end CI/CD pipelines (GitLab CI/CD · Docker · AWS EC2), and achieved ≥70% unit-test coverage (JUnit 5 / Mockito / JaCoCo). IELTS 6.0 · Agile/Scrum · Strong OOP & SOLID foundations · Eager to contribute to enterprise environments.

TECHNICAL SKILLS

Languages

Java 17, Python 3.x, JavaScript (ES6+), SQL

Frameworks

Spring Boot 3.x, Spring Security, Spring Data JPA, FastAPI, React 18, Redux Toolkit

Databases

MySQL 8, PostgreSQL 16, Redis 7, TiDB Cloud

Cloud & DevOps

AWS (EC2, RDS, S3), Docker, GitLab CI/CD, GitHub Actions

Testing & QA

JUnit 5, Mockito, Pytest, JaCoCo (≥70%), Swagger

Tools & Practices

Git, Flyway, HikariCP, JWT, Jira, Agile/Scrum, SOLID

EDUCATION

University of Information Technology - VNU-HCM (UIT)

2023 – 2027

B.Sc. Information Systems · GPA: 8.0 / 10.0

Relevant Coursework: Database Management Systems, Software Engineering, OOP & Design Patterns, Software Project Management (Agile/Scrum).

English: IELTS 6.0 — Professional Working Proficiency · Vietnamese: Native

KEY PROJECTS

Concert-Booking — High-Concurrency Ticket Booking Platform

Jan 2026 – Present

Personal Project · gitlab.com/tuanwannafly1/concert-booking

Stack: Java 17 · Spring Boot 3.2 · MySQL 8 · Redis 7 · React · JUnit 5 · Docker · GitLab CI/CD · AWS EC2

- Architecture & API Integration:** Architected a Spring Boot monolith handling ~500 req/min peak load; exposed RESTful APIs consumed by a React SPA with JWT-secured authentication.
- Concurrency & Overselling Prevention:** Engineered atomic ticket deduction combining Redis DECRBY with JPA optimistic locking, preventing overselling under concurrent flash-sale traffic.
- Idempotency Middleware:** Built X-Idempotency-Key middleware (Redis-backed) enforcing exactly-once booking semantics.
- CI/CD Pipeline:** Automated Maven build → JaCoCo coverage gating → Docker-in-Docker build → SSH deployment to AWS EC2 via GitLab CI/CD.

JobMarket — Full-Stack Recruitment Platform

Dec 2025 – Present

Personal Project · gitlab.com/tuanwannafly1/jobmarket

Stack: Java 17 · Spring Boot 3 · React 18 · TiDB Cloud · FastAPI · Groq LLM · AWS (EC2, S3) · Docker · CI/CD

- Backend Architecture & RBAC:** Designed Spring Boot REST backend with custom RBAC interceptor and SpringFilter JPA query optimization.
- AI Resume Parsing:** Built a FastAPI microservice integrating Groq LLM to parse PDF/DOCX resumes and extract ATS match scores.
- Frontend Resilience:** Resolved concurrent 401 token-refresh race conditions on Axios interceptors using asyncmutex locks.

SmartRestaurant — Enterprise Restaurant Management System

Jan 2026 – Present

Main Developer · github.com/tuanwannafly1/restaurant

Stack: Java 17 · PostgreSQL 16 · Redis · WebSockets · Google Gemini 2.0 AI · Docker · JUnit 5

- Distributed Locking & Concurrency:** Implemented two-tier locking (Redis distributed locks + conditional SQL UPDATES) to eliminate race conditions.
- Database Migration:** Led full Oracle → PostgreSQL 16 migration; tuned HikariCP connection pool and transaction isolation levels.
- AI & External API Integration:** Integrated Google Gemini 2.0 Flash API to automate intelligent menu description generation.
- Real-Time Architecture:** Designed WebSocket pub/sub event system to synchronize cross-client events (kitchen ↔ waiter ↔ cashier) in real time.

EXPERIENCE

Cortex Linux Ecosystem — Open Source Contributor (Project-based)

Dec 2025 – Feb 2026

Python · Linux · NVIDIA PRIME Offload · Pytest

Engineered a Python-based Hybrid GPU Manager implementing per-application hardware routing via environment variables to optimize GPU power utilization on dual-GPU Linux systems. Authored a comprehensive Pytest test suite validating GPU state detection and battery estimation heuristics (100% pass rate). Managed full Git contribution lifecycle following upstream project contribution guidelines.

AWARDS & RECOGNITION

Top 30 Finalist · The Spark Programming Challenge *May 2026*

Top 5 Finalist Teams · The Connection Talent Network *May 2026*